

## 1.1 Meaning of research and educational research

**Research** refers to the systematic collection and analysis of data related to a specific field or topic. It involves rigorous investigation and empirical methods to explore, understand, and contribute to knowledge in that area.

Researchers aim to discover patterns, relationships, and underlying principles by examining data from various sources. The process of research often follows specific procedures and emphasizes accuracy, objectivity, and validation.

## What is Educational Research?

Educational research is a type of systematic investigation that applies empirical methods to solving challenges in education. It adopts rigorous and well-defined scientific processes in order to gather and analyze data for problem-solving and knowledge advancement.

[J. W. Best defines educational research](#) as that activity that is directed towards the development of a science of behavior in educational situations. The ultimate aim of such a science is to provide knowledge that will permit the educator to achieve his goals through the most effective methods.

The primary purpose of educational research is to expand the existing body of knowledge by providing solutions to different problems in pedagogy while improving teaching and learning practices. Educational researchers also seek answers to questions bothering on learner motivation, development, and classroom management.

## Characteristics of Education Research

While educational research can take numerous forms and approaches, several characteristics define its process and approach. Some of them are listed below:

1. It sets out to solve a specific problem.
2. Educational research adopts [primary and secondary research methods](#) in its [data collection process](#). This means that in educational research, the investigator relies on first-hand sources of information and [secondary data](#) to arrive at a suitable conclusion.
3. Educational research relies on [empirical evidence](#). This results from its largely scientific approach.

4. Educational research is objective and accurate because it measures verifiable information.
5. In educational research, the researcher adopts specific methodologies, detailed procedures, and analysis to arrive at the most objective responses
6. Educational research findings are useful in the development of principles and theories that provide better insights into pressing issues.
7. This research approach combines [structured, semi-structured, and unstructured questions](#) to gather verifiable data from respondents.
8. Many educational research findings are documented for peer review before their presentation.
9. Educational research is interdisciplinary in nature because it draws from different fields and studies complex factual relations.

1.2 Research issues and problems in educational field

1.3 Steps in educational research

### 1. Selecting a Problem:

- **Identifying Area of Interest:** Start by identifying a broad area of interest within your discipline.
- **Gap Analysis:** Conduct a literature review to identify gaps, controversies, or unanswered questions in the existing research.
- **Refining the Problem:** Narrow down your focus to a specific research problem that is clear, concise, and researchable.

*Example:* Suppose you are interested in the field of education. You observe a lack of research on the impact of online learning environments on the academic performance of primary school students, particularly in underprivileged areas.

### 2. Reviewing the Literature:

- **Search Strategy:** Develop a systematic approach to searching and reviewing relevant literature, including databases, journals, and other sources.
- **Organizing Information:** Categorize and organize the literature based on themes, methodologies, and key findings.
- **Identifying Trends:** Summarize and synthesize the current state of knowledge, highlighting key trends and debates.

*Example:* After conducting a literature review, you find existing studies on online learning but notice a gap in research specifically focused on the socio-economic factors

influencing the effectiveness of online education for primary school students in underprivileged communities.

### 3. Designing the Research:

- **Research Design:** Choose between qualitative, quantitative, or mixed-methods research based on your research question (Dahal & Subedi, 2023).
- **Sampling Strategy:** Determine the target population and select a representative sample using appropriate sampling techniques.
- **Data Collection Instruments:** Develop tools such as surveys, questionnaires, interviews, or observation protocols.
- **Ethical Considerations:** Address ethical concerns and obtain necessary approvals from institutional review boards (IRBs).

*Example:* You decide on a mixed-methods approach. For the quantitative part, you plan to administer standardized tests to assess academic performance. Qualitatively, you intend to conduct interviews with students, parents, and teachers to understand the contextual factors affecting learning outcomes.

### 4. Collecting the Data:

- **Pilot Testing:** Conduct a small-scale test of your data collection instruments to identify and rectify any issues.
- **Data Collection Procedures:** Implement the finalized data collection plan, ensuring consistency and accuracy.
- **Data Recording:** Systematically record and organize the collected data for later analysis.

*Example:* You select a representative sample of primary schools from underprivileged areas. You administer standardized tests to students and conduct interviews with parents, teachers, and students. Ethical considerations include obtaining consent from all participants and ensuring the confidentiality of the collected data.

### 5. Analyzing the Data:

- **Statistical Analysis:** If using quantitative methods, employ statistical techniques relevant to your study.
- **Qualitative Analysis:** If using qualitative methods, apply coding, thematic analysis, or other relevant approaches.
- **Data Quality Control:** Check and clean the data to ensure accuracy and reliability.
- **Interpretation:** Provide a detailed interpretation of the results, considering both statistical significance and practical significance.

- *Example:* Using statistical analysis, you compare the academic performance scores of students in online learning environments with those in traditional classrooms. Simultaneously, you conduct thematic analysis on the qualitative data from interviews to identify key themes related to socio-economic factors affecting learning.

## 6. Interpreting the Findings:

- **Contextualization:** Place your findings in the context of existing literature and theoretical frameworks.
- **Discussion of Patterns and Trends:** Discuss any patterns or trends observed in the data.
- **Alternative Explanations:** Consider and discuss alternative explanations for your findings.
- **Limitations:** Acknowledge and discuss the limitations of your study.
- *Example:* Your analysis reveals that while online learning can be effective, access to technology and parental involvement significantly influence the outcomes. The thematic analysis uncovers nuances in how economic disparities impact the ability of students to engage with online education.

## 7. Drawing Conclusions:

- **Summarize Key Findings:** Clearly state the main findings of your study.
- **Implications:** Discuss the implications of your results for theory, practice, or policy.
- **Recommendations:** Provide recommendations for future research in the field.
- **Conclusion:** Sum up the research process, emphasizing the contributions and significance of your study.
- *Example:* You conclude that online learning can be a viable option for underprivileged primary school students, but addressing socio-economic factors is crucial for its success. Your findings suggest the need for targeted interventions to bridge the digital divide and enhance parental support for students in such environments.

## References

Dahal, S., & Subedi, S. (2023). Drug addiction . *Ashoj* .